

Day 52 - LAN Architectures

Wednesday, September 9, 2026

8:37 PM

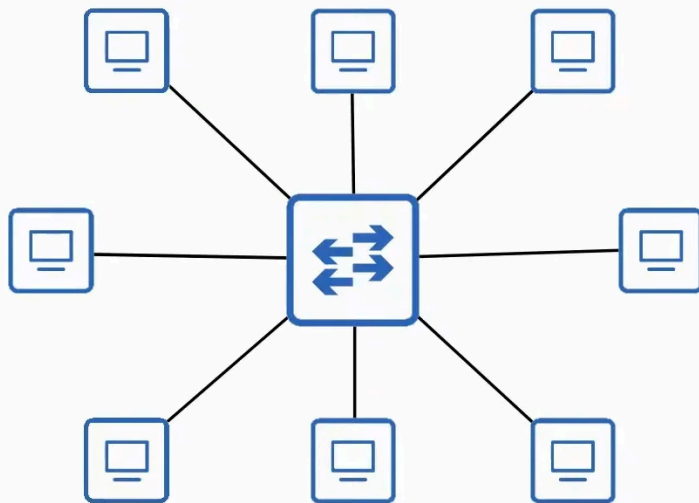
LAN Architecture Info

In order to better troubleshoot and configure networks, it's important to know some basics of network designing.

Common Terminologies

1. Star

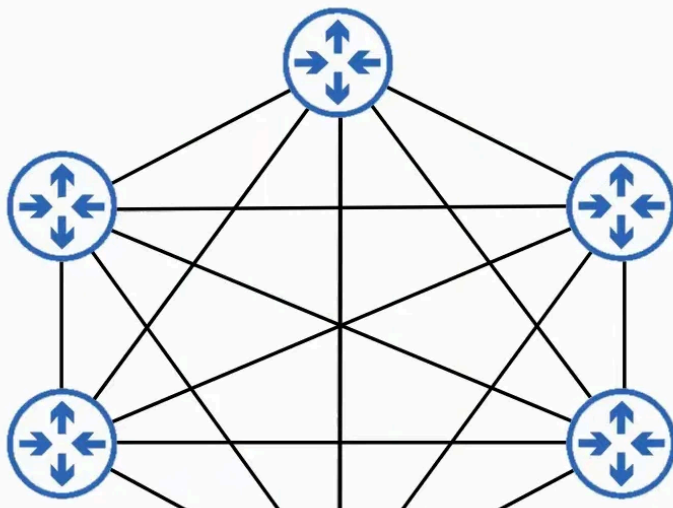
A **star** topology is formed when several devices all connect to one central device. They can be drawn in a 'star' shape, so they're often called 'star topologies'

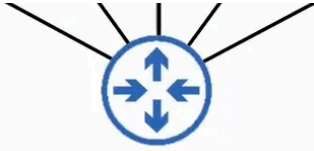


Star topology

2. Full Mesh

A **Full Mesh** topology is formed when each device is connected to each other device.

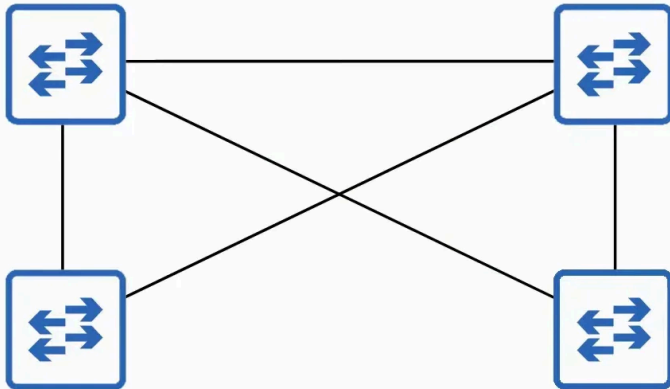




Full Mesh topology

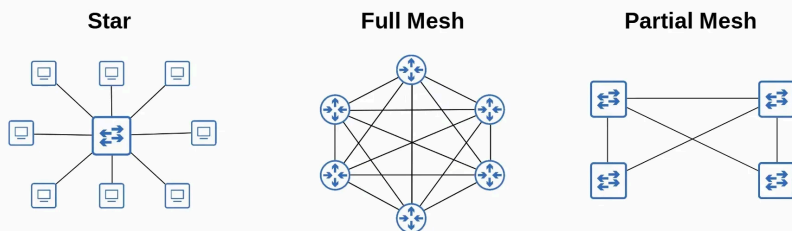
3. Partial Mesh

A **Partial Mesh** topology is formed when some devices are connected to each other, but not all.



Partial Mesh topology

Comparison



Comparison between all 3 topologies

Architecture Designs

Two-Tier Campus LAN Design

The two-tier LAN design is a design usually used in smaller scale networks. It consists of two hierarchical layers:

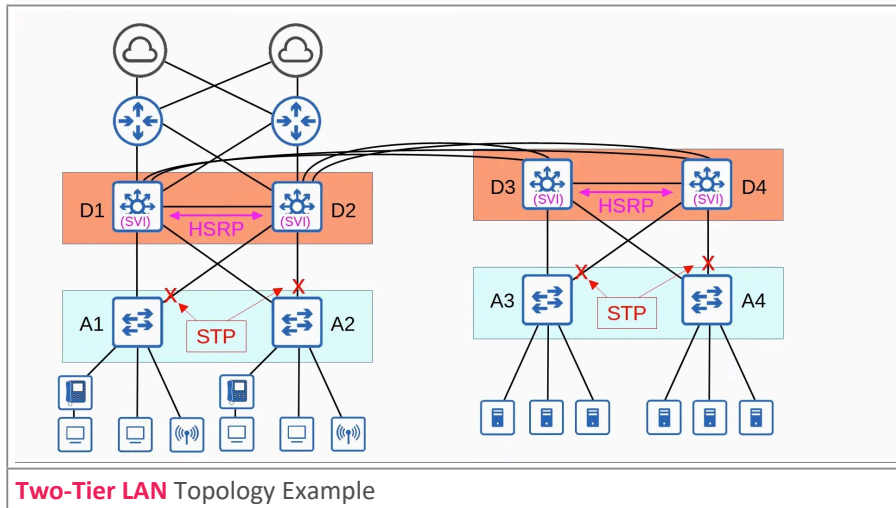
1. Access Layer

- The layer that end hosts connect to. (PCs, phones, wireless APs, printers, cameras, and etc.)
- Typically has switches that have a lot of ports for end hosts to connect to.
- **QoS** marking is typically done here.
- Security services like **port security**, **DHCP snooping**, **DAI**, etc are typically performed here.
- Switchports might be **PoE**-enabled for wireless APs, **IP phones**, and etc.

2. Distribution Layer

- Sometimes referred to as the '**Core-Distribution Layer**'.
- Aggregates connections from the **Access Layer** switches.
- Typically acts as the border between **Layer 2** and **Layer 3**.
- Connections between **Distribution Layer** switches are **Layer 3**

- Connections between **Distribution Layer** switches are **Layer 3**.
- In **two-tier LANs**, it connects to services such as the internet, WAN, etc.
- This design is also referred to as the “**Collapsed Core**” design, because it omits a layer that is found in the **three-tier LAN** design called the **Core Layer**.
- Connections between the **Access Layer** and the **Distribution Layer** are **Layer 2**, which can lead to **broadcast storms**, so **STP** is activated in order to disable some interfaces and prevent them from happening.
- Some extra **Distribution Layer** switches are deployed with a **FHRP** protocol in order to provide redundancy.



In the previous image, you can spot the three previously mentioned topologies.

- There are **star** topologies present where the end hosts connect to the **Access Layer** switches.
- There are **partial mesh** topologies present where the **Access Layer** switches connect with the **Distribution Layer** switches.
- There are **full mesh** topologies present where the **Distribution Layer** switches all connect to each other.

Distribution Layer: Layer 3 connections

- Connections between **Distribution Layer** switches are **Layer 3**.
- This allows for routing information to be shared via dynamic routing protocols, such as **OSPF**, for example.

Three-Tier Campus LAN Design

In large LAN networks with many **Distribution Layer** switches, the number of connections required between **Distribution Layer** switches grows rapidly. This makes it more difficult to scale networks and make them larger.

To help scale large LAN networks, you can add a **Core Layer**. Which turns the design into a three-tier LAN. (Cisco recommends adding a **Core Layer** if there are more than three **Distribution Layers** in a single location.)

It consists of:

1. Access Layer

- The layer that end hosts connect to. (PCs, phones, wireless APs, printers, cameras, and etc.)
- Typically has switches that have a lot of ports for end hosts to connect to.
- **QoS** marking is typically done here.
- Security services like **port security**, **DHCP snooping**, **DAI**, etc are typically performed here.
- Switchports might be **PoE**-enabled for wireless APs, **IP phones**, and etc.

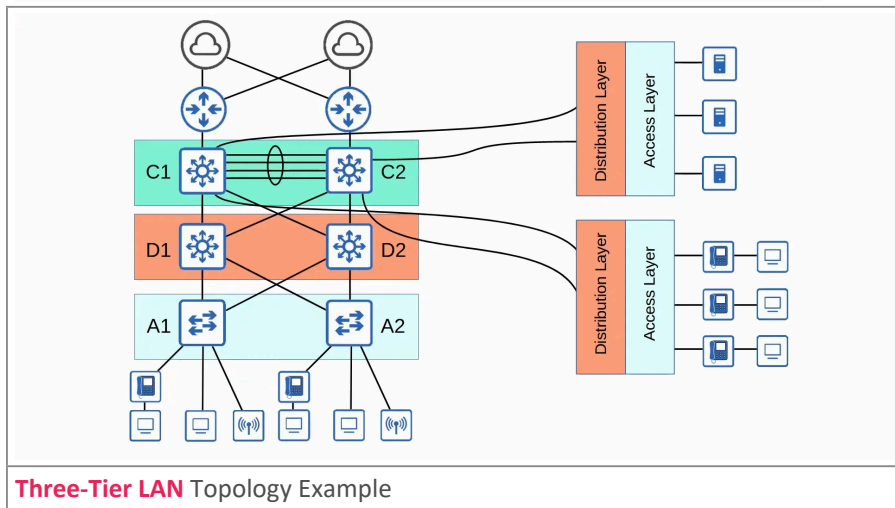
2. Distribution Layer

- Sometimes referred to as the ‘**Core-Distribution Layer**’.
- Aggregates connections from the **Access Layer** switches.
- Typically acts as the border between **Layer 2** and **Layer 3**.
- Connections between **Distribution Layer** switches are **Layer 3**.

3. Core Layer

- Connects **Distribution Layers** together in large LAN networks.
- Connections are all **Layer 3**, therefore, there is no **STP**.
- It is focused on speed (‘fast transport’)
- CPU intensive operations such as security, **QoS** marking/**classification**, security features and etc. should be avoided at this layer.

- Should maintain connectivity throughout the LAN even if the devices fail.



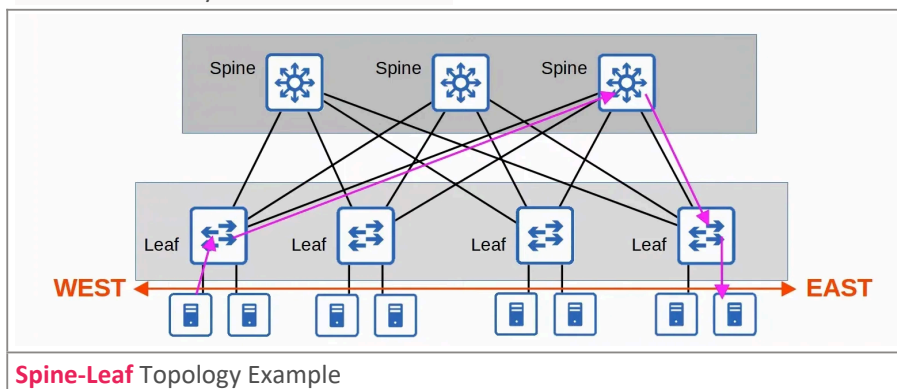
Spine-Leaf Architecture

Spine-Leaf Architecture Reasoning

- Data centers are dedicated spaces/buildings used to store computer systems such as servers and network devices.
- Traditional data centers used a **three-tier LAN** architecture. Which worked well when most of the traffic in the data center was *North-South*.
- With the precedence of virtual servers, applications are often deployed in a distributed manner across multiple physical servers, which increases the amount of *East-West* traffic in the data center.
- The traditional **three-tier LAN** led to bottlenecks as well as variability in latency.
- To solve this issue, **Spine-Leaf** architecture (**Clo's** architecture) has become prominent in data centers.

Spine-Leaf Architecture Details

- Every **Leaf** switch is connected to every **Spine** switch
- Every **Spine** switch is connected to every **Leaf** switch
- **Leaf** switches do not connect to other **Leaf** switches.
- **Spine** switches do not connect to other **Spine** switches.
- End hosts only connect to **Leaf** switches.
- The path taken by traffic is randomly chosen to balance the traffic load among the **Spine** switches.
- Each server is separated by the same number of 'hops' (except those connected to the same **Leaf**), providing a consistent latency for *East-West* traffic.



Small Office / Home Office (SOHO) Networks

Small Office / Home Office (SOHO) refers to the office of a small company, or a small home office with few devices.

- **SOHO** networks don't have complex needs, so all networking functions are usually provided by a single device, often called a 'home router' or 'wireless router' which can serve as a:
 1. *Router*

2. Switch
3. Firewall
4. Wireless Access Point
5. Modem

